

SERTOLI CELLS

- **Definition** : Specialized supporting cells for spermatogenic cells within the seminiferous tubules.
- **Synonyms** : Also known as **Sustentacular cells** or **Nurse cells**.
- **Morphology & Location** :

Large & tall, irregular columnar cells



Extend from the basement membrane to the lumen of the seminiferous tubule

- **Attachment** : Germ cells in the seminiferous tubule attach to Sertoli cells via **cytoplasmic connections**, which persist until fully mature spermatozoa are released (*spermiation*).

FUNCTIONS OF SERTOLI CELLS

Function / Secretion	Physiological Role & Mechanism
1. Support & Nourishment	Provide structural support, protection, and vital nutrients to developing spermatogenic cells in the seminiferous tubules.
2. Aromatase Enzyme Secretion	Converts androgens into estrogen : $\text{Androgens} \xrightarrow{\text{Aromatase}} \text{Estrogen}$
3. Androgen-Binding Protein (ABP)	Binds to testosterone, maintaining high local androgen concentration essential for testosterone activity during spermatogenesis .
4. Estrogen-Binding Protein (EBP)	Facilitates estrogen transport and local action within the seminiferous epithelium.
5. Inhibin Secretion	Exerts negative feedback to inhibit FSH secretion from the anterior pituitary gland.

Function / Secretion	Physiological Role & Mechanism
6. Activin Secretion	Stimulates and ↑ increases FSH release from the anterior pituitary gland.
7. Müllerian Regression Factor (MRF) <i>(Müllerian Inhibiting Substance / MIS)</i>	Secreted in fetal testes; responsible for the regression of the Müllerian (paramesonephric) duct during male fetal sex differentiation.